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How the Code Works

This project implements a preemptive multithreading package for the 8051 microcontroller. Unlike the cooperative version, threads do not need to manually call `ThreadYield()` to give up control. Instead, Timer 0 is configured in `Bootstrap()` to generate periodic interrupts. Each Timer 0 interrupt jumps to `myTimer0Handler()`, which performs the context switch automatically.

The thread package uses manually allocated internal RAM to store the saved stack pointer for each thread, the current thread ID, and the active-thread bitmap. Each thread has its own 16-byte stack region and uses a separate 8051 register bank. Thread 0 uses register bank 0, thread 1 uses register bank 1, and so on. This allows the program to avoid saving and restoring registers R0 through R7 for every thread.

When `Bootstrap()` runs, it initializes the thread-management variables, creates `main()` as the first thread, enables the Timer 0 interrupt, starts Timer 0, and restores the context of the first thread. After this, `main()` creates the Producer thread and then runs the Consumer function.

`ThreadCreate()` is responsible for creating a new thread. It finds an unused thread ID, marks it as active in the bitmap, assigns the correct stack region, and pushes the starting function address onto the new stack. It also initializes the saved register context, including ACC, B, DPL, DPH, R0, and PSW. The PSW value selects the register bank assigned to the thread. After the initial stack is prepared, the stack pointer is saved in the `savedSP` array.

Context switching is handled by `myTimer0Handler()`. When Timer 0 interrupts the current thread, the handler saves the current thread's registers and stack pointer. It then selects the next valid thread using a round-robin policy by checking the thread bitmap. After selecting the next thread, it restores that thread's saved stack pointer and registers, then returns from the interrupt using `RETI`. This causes execution to continue in the selected thread instead of the interrupted one.

The test program uses a single-buffer producer-consumer example. The Producer generates characters from `A` to `Z` repeatedly and writes one character at a time into `sharedBuffer`. The Consumer waits until the buffer is full, reads the character, sends it to the UART through `SBUF`, and marks the buffer as empty again. The variable `bufferFull` is used to indicate whether the shared buffer currently contains valid data.

Because the Producer and Consumer share `sharedBuffer` and `bufferFull`, interrupts are temporarily disabled using `EA = 0` while these shared variables are being checked or updated.

This prevents Timer 0 from preempting a thread in the middle of a shared-buffer operation. After the critical section is complete, interrupts are re-enabled with `EA = 1`.

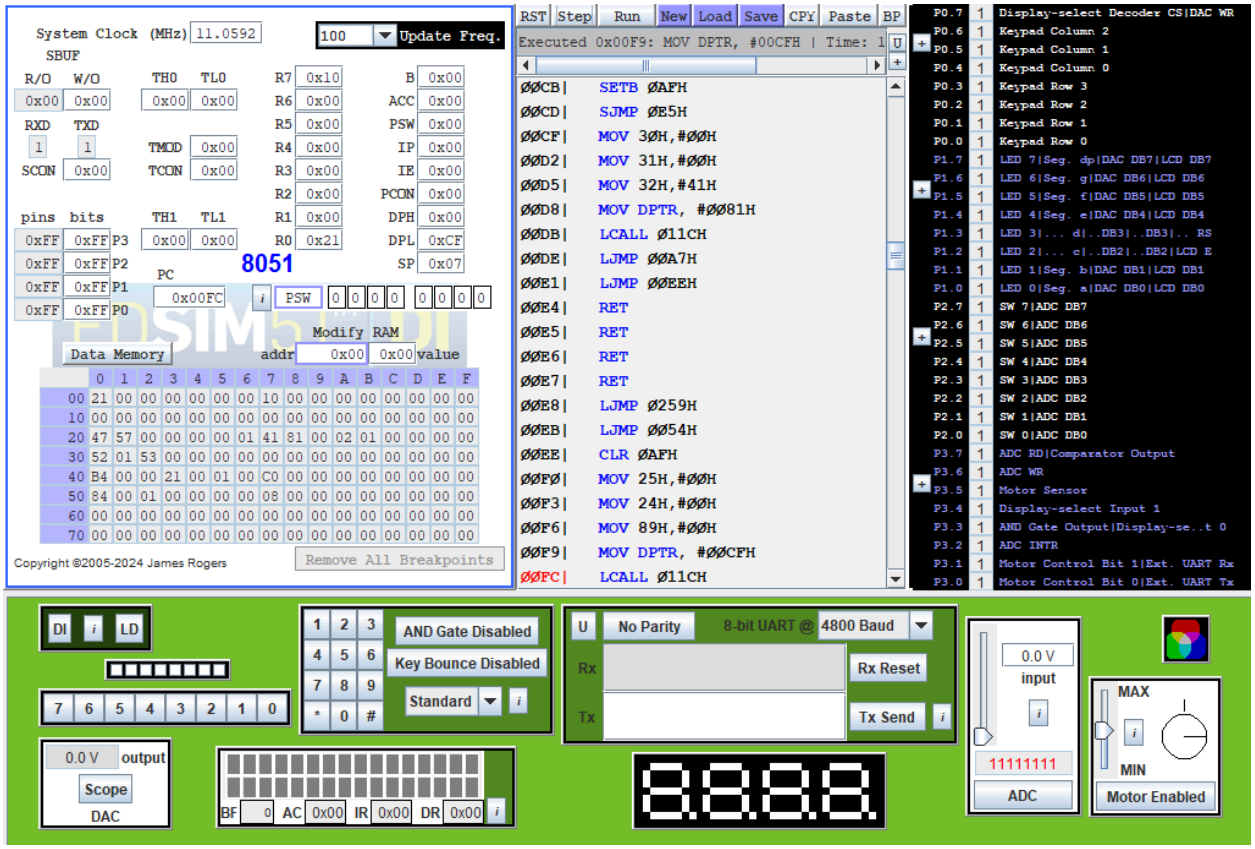
When the program runs correctly in EdSim51, the UART receive window displays repeating letters from `A` to `Z`. This shows that the Producer is generating characters, the Consumer is transmitting them, and Timer 0 is preemptively switching between the two threads.

3.1 Screenshots for compilation

```
PS C:\Users\Theresia\Downloads\112006221_ppc2> make clean
powershell -NoProfile -Command "Remove-Item -ErrorAction SilentlyContinue *.h
ex,*.ihx,*.lnk,*.lst,*.map,*.mem,*.rel,*.rst,*.sym,*.asm,*.lk"
PS C:\Users\Theresia\Downloads\112006221_ppc2> make
sdcc -c testpreempt.c
sdcc -c preemptive.c
sdcc  -o testpreempt.hex testpreempt.rel preemptive.rel
PS C:\Users\Theresia\Downloads\112006221_ppc2> Get-ChildItem


    Directory: C:\Users\Theresia\Downloads\112006221_ppc2


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         6/13/2026   6:25 AM            559 Makefile
-a----         6/13/2026   6:51 AM          14658 preemptive.asm
-a----         6/13/2026   6:58 AM           7396 preemptive.c
-a----         6/13/2026   6:20 AM            289 preemptive.h
-a----         6/13/2026   6:51 AM          40164 preemptive.lst
-a----         6/13/2026   6:51 AM           6265 preemptive.rel
-a----         6/13/2026   6:51 AM          40818 preemptive.rst
-a----         6/13/2026   6:51 AM          40876 preemptive.sym
-a----         6/13/2026   6:51 AM          14125 testpreempt.asm
-a----         6/13/2026   6:48 AM           5001 testpreempt.c
-a----         6/13/2026   6:51 AM           1735 testpreempt.hex
-a----         6/13/2026   6:51 AM            261 testpreempt.lk
-a----         6/13/2026   6:51 AM          35887 testpreempt.lst
-a----         6/13/2026   6:51 AM          17343 testpreempt.map
-a----         6/13/2026   6:51 AM           1192 testpreempt.mem
-a----         6/13/2026   6:51 AM           5540 testpreempt.rel
-a----         6/13/2026   6:51 AM          36445 testpreempt.rst
-a----         6/13/2026   6:51 AM          40315 testpreempt.sym


PS C:\Users\Theresia\Downloads\112006221_ppc2>
```

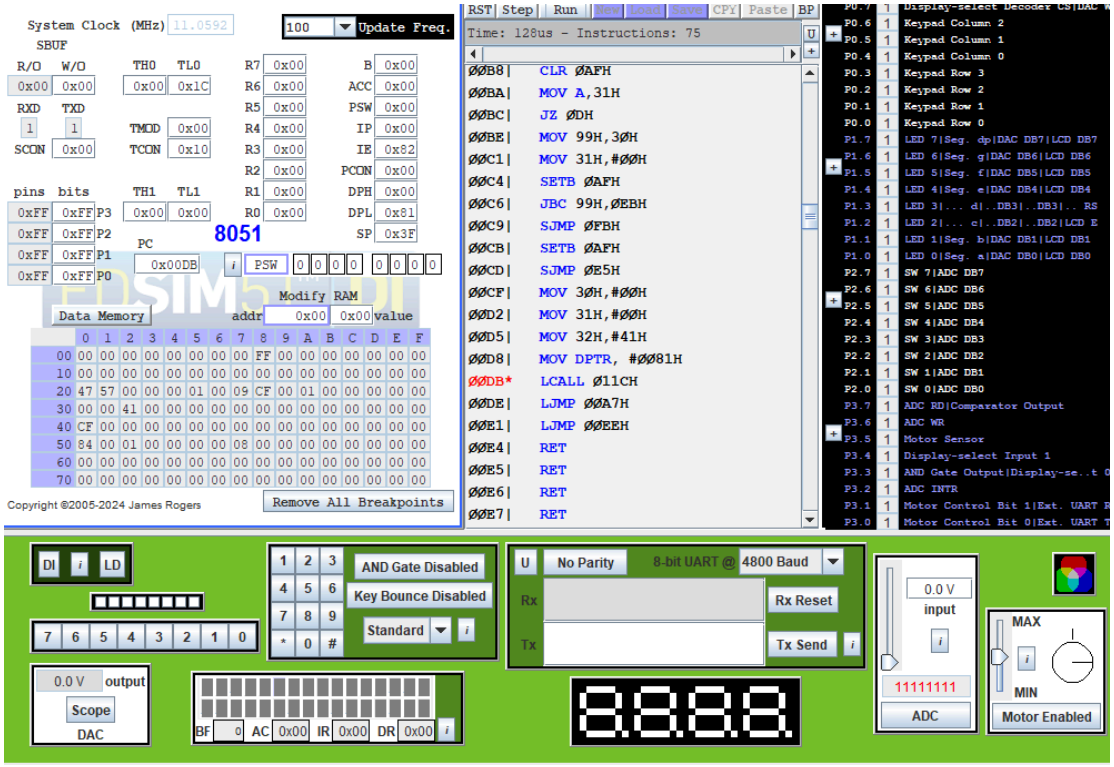
3.2

Part 1: Before ThreadCreate call



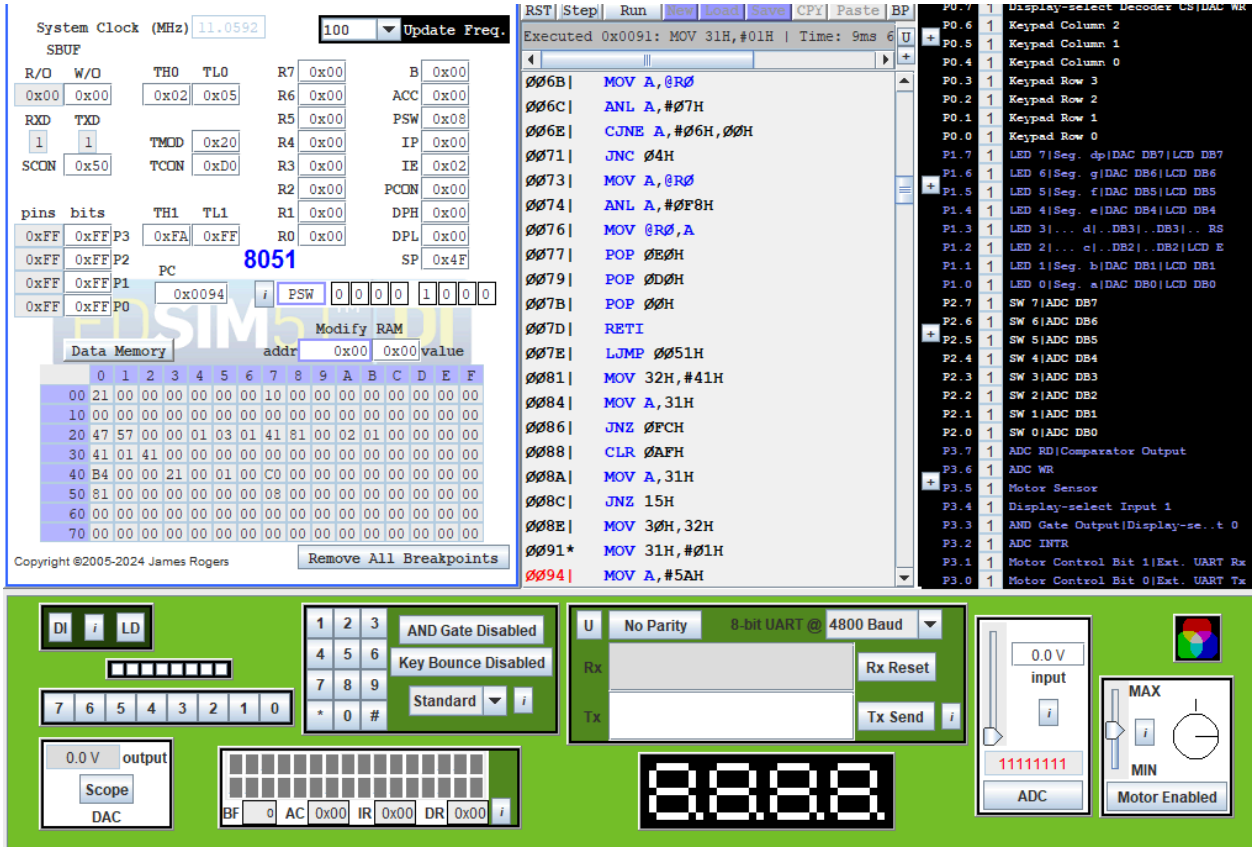
This screenshot was taken immediately before Bootstrap() calls ThreadCreate(main). At this point, no thread has been created yet, so threadBitmap at address 0x25 is 0x00, and currentThread at address 0x24 is 0x00. The stack pointer is still at 0x07, which is the default startup stack location. During ThreadCreate(main), thread 0 is allocated, a new stack is prepared for it starting at address 0x40, the address of main() is pushed onto the stack, and the initial register context is added. The final stack pointer is then stored in savedSP[0], and the thread bitmap becomes 0x01.

Part 2: Before ThreadCreate(Producer)



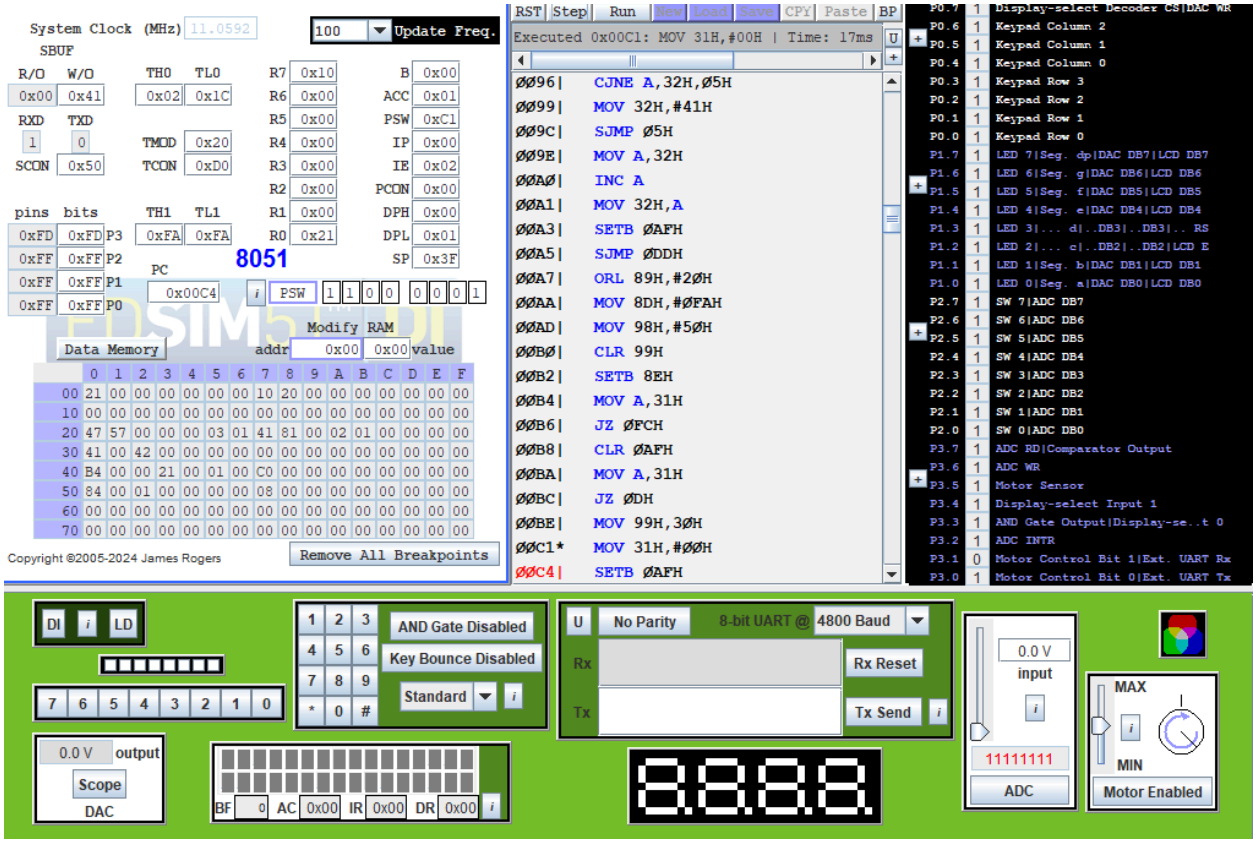
This screenshot was taken immediately before main() calls ThreadCreate(Producer). At this point, thread 0 has already been created and is running main(), so currentThread at address 0x24 is 0x00. The thread bitmap at address 0x25 is 0x01, showing that only thread 0 is currently active. DPTR contains 0x0081, which is the address of Producer() and is passed to ThreadCreate() as the function pointer. During this call, thread ID 1 is allocated and the bitmap becomes 0x03. A new stack is prepared for thread 1 beginning at address 0x50. The address of Producer() and the initial register context are pushed onto this stack. The PSW value selects register bank 1, and the final stack pointer is saved in savedSP[1].

Part 3: Producer Running



The Producer is running because the program counter is inside the Producer() function and currentThread at address 0x24 contains 0x01, meaning thread 1 is active. The PSW value is 0x08, which selects register bank 1 for thread 1. The shared buffer at address 0x30 contains 0x41, representing the character 'A', and bufferFull at address 0x31 is 0x01. This confirms that the Producer has generated a character and placed it into the shared buffer. Since this is the preemptive version, the Producer does not call ThreadYield(); Timer 0 performs the context switching automatically.

Part 4: Consumer Running



The Consumer is running because the program counter is inside the Consumer() function, near the instructions that send the shared character to the UART and clear the buffer flag. The current thread ID at address 0x24 is 0x00, indicating that thread 0 is active. The value in sharedBuffer at address 0x30 is 0x41, which represents 'A', and the same value has been copied into SBUF. The value of bufferFull at address 0x31 is now 0x00, showing that the Consumer has consumed the character and released the shared buffer for the Producer. Since this is the preemptive version, Timer 0 handles the switching between Producer and Consumer automatically.

Part 5: How can you tell that the interrupt is triggering on a regular basis?

The Timer 0 interrupt is triggered on a regular basis because the program continues switching between the Producer and Consumer even though neither function calls ThreadYield(). In EdSim51, Timer 0 is enabled because IE = 0x82, and Timer 0 is running as shown by the changing TH0 and TL0 values. The value of currentThread at address 0x24 changes between 0x00 and 0x01, showing that the interrupt handler is periodically saving the current thread state and restoring the next thread. The UART output also continues to print characters, confirming that both threads are being scheduled repeatedly by the timer interrupt.